Code in Python

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# Hash Table using Division Method and Linear Probing

SIZE = 10
DELETED = "DELETED"
table = [None] * SIZE


# Hash function (division method)
def hash_function(key):
    return key % SIZE


# Insert a key
def insert(key):
    index = hash_function(key)
    for i in range(SIZE):
        pos = (index + i) % SIZE
        if table[pos] is None or table[pos] == DELETED:
            table[pos] = key
            print("Key inserted at index", pos)
            return
    print("Hash table is full.")


# Search for a key
def search(key):
    index = hash_function(key)
    for i in range(SIZE):
        pos = (index + i) % SIZE
        if table[pos] == key:
            print("Key found at index", pos)
            return
        if table[pos] is None:
            break
    print("Key not found.")


# Delete a key
def delete(key):
    index = hash_function(key)
    for i in range(SIZE):
        pos = (index + i) % SIZE
        if table[pos] == key:
            table[pos] = DELETED
            print("Key deleted.")
            return
        if table[pos] is None:
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            break
    print("Key not found.")


# Display the table
def display():
    print("\nIndex\tKey")
    print("-" * 15)
    for i in range(SIZE):
        if table[i] is None or table[i] == DELETED:
            print(i, "\t", "Empty")
        else:
            print(i, "\t", table[i])


# Main program
while True:
    print("\n--- Hash Table Menu ---")
    print("1. Insert")
    print("2. Search")
    print("3. Delete")
    print("4. Display")
    print("5. Exit")
    choice = int(input("Enter your choice: "))

    if choice == 1:
        insert(int(input("Enter key to insert: ")))
    elif choice == 2:
        search(int(input("Enter key to search: ")))
    elif choice == 3:
        delete(int(input("Enter key to delete: ")))
    elif choice == 4:
        display()
    elif choice == 5:
        print("Program terminated.")
        break
    else:
        print("Invalid choice!")
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